# Patrick Soga

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## Education

**University of Virginia** — Aug 2023 – Present
*Ph.D. in Computer Science – 3.93/4.0. Advisor: Prof. Jundong Li.* — *Charlottesville, VA*

**University of Notre Dame** — Aug 2018 – Jan 2023
*B.S. Computer Science, B.A. Philosophy – 3.85/4.0, Cum Laude.* — *Notre Dame, IN*

## Publications

* indicates equal contribution.

1. Zhenyu Lei, **Patrick Soga**, Yaochen Zhu, Yinhan He, Yushun Dong, and Jundong Li. MolEdit: Knowledge Editing for Multimodal Molecule Language Models. In *WSDM 2026.*
2. **Patrick Soga**, Zhenyu Lei, Yinhan He, Camille Bilodeau, and Jundong Li. Energy-Based Models for Predicting Mutational Effects on Proteins. In *KDD 2025.*
3. **Patrick Soga**, Yushun Dong, Yaochen Zhu, Jundong Li, Tong Zhao, and Neil Shah. VirtualGCN – Enhancing Graph Collaborative Filtering with Virtual Interactions. In *IEEE Big Data 2025.*
4. Yushun Dong, **Patrick Soga**, Yinhan He, Song Wang, and Jundong Li. Graph Neural Networks Are More Than Filters: Revisiting and Benchmarking from A Spectral Perspective. In *ICLR 2025.*
5. **Patrick Soga**, Zhenyu Lei, Camille L. Bilodeau, and Jundong Li. Deep Interactions for Multimodal Molecular Property Prediction. In *PAKDD 2025.*
6. Yinhan He, Zaiyi Zheng, **Patrick Soga**, Yaochen Zhu, Yushun Dong, and Jundong Li. Explaining Graph Neural Networks with Large Language Models: A Counterfactual Perspective on Molecule Graphs. In *EMNLP 2024 (Findings).*
7. Jennifer J. Schnur*, Angélica Garcia-Martínez*, **Patrick Soga***, Karla Badillo-Urquiola*, et al. SaludConectaMX: Lessons Learned from Deploying a Cooperative Mobile Health System for Pediatric Cancer Care in Mexico. In *CSCW 2024.*
8. **Patrick Soga** and David Chiang. Bridging Graph Position Encodings for Transformers with Weighted Graph-Walking Automata. In *TMLR, 2023.*
9. Steven Krieg, William Burgis, **Patrick Soga**, and Nitesh Chawla. Deep Ensembles for Graphs with Higher-order Dependencies. In *ICLR 2023.*

## In Submission

1. **Patrick Soga**, Camille Bilodeau, and Jundong Li. Ensemble-IF: Towards Protein Design for Target Conformational Populations. In *Submission 2026.*
2. **Patrick Soga**, Zhenyu Lei, Yinhan He, Camille Bilodeau, and Jundong Li. GuideDock: Geometry-Guided Diffusion for Rigid Protein-Protein Docking. In *Submission 2026.*

## Work Experience

**AT&T** — August 2026 – October 2026
*Research Intern* — *Bedminster, NJ*

- Developing flow-matching algorithms with learnable transport routes and schedules for coordinating complex geometric transformations.
- Achieving state-of-the-art protein-ligand docking RMSD and applying it to internal business settings.

**AbbVie** — May 2026 – August 2026
*Graph Machine Learning Intern* — *North Chicago, IL*

- Developed knowledge graph neural network models for predicting clinical trial outcomes.
- Increased AUROC by up to 4% overall and 7% on zero-history trials.

**Lucy Family Institute for Data and Society** June 2021 – May 2022
*Software Developer* *Notre Dame, IN*

- Built web (React) and mobile (Flutter) apps for gathering and managing patient medical information and assessing cancer patient risk for HIMFG, a premier hospital in Mexico City, Mexico.
- Currently deployed with >100 users and records. Resulted in [7].

**FloVision Solutions** July 2021 – March 2022
*ML & Software Engineer* *Remote*

- Wrote Google Cloud Function pipelines for automating annotations and deployed inference jobs with Docker on Google Cloud Compute Engine VMs.
- Tuned CNN architectures using transfer learning techniques for food image classification for waste reduction.

## Research Experience

**Snap Collaboration** February 2024 – October 2024

- Developed a novel GCN-inspired matrix factorization method for graph collaborative filtering-based recommendation. Supervised by research scientists from Snap. Resulted in [3].

## Teaching

- TA for CS 6316 Machine Learning, Fall 2025.
- TA for CS 4774 Machine Learning, Spring 2025 and 2026.

## Academic Service

- Reviewer for NeurIPS 2026, ICML 2025-2026, ICLR 2026-2027, AAAI 2027, CIKM 2026, WWW 2025, and TMLR.
- External reviewer for KDD 2024-2025, IJCAI 2024, WSDM 2024, and DSAA 2024.

## Skills

**Programming & Frameworks:** Python, PyTorch, PyTorch Geometric, Deep Graph Library (DGL), RDKit, Neo4j, Biopython, Biotite, C, JavaScript, TypeScript, PostgreSQL, ReactJS, Angular, NodeJS, Flask.

**Software & Tools:** Git, Ubuntu, SLURM, Google Cloud Platform (Cloud Functions, Compute Engine, Firestore), AWS (S3, Lambda, EC2), PyMol, AlphaFold & Boltz co-folding models, ProteinMPNN.

## Awards & Distinctions

- ICML 2026 Gold Reviewer (top-25% of reviewers)
- Student Travel Award: KDD 2025, PAKDD 2025
- Phi Beta Kappa, Spring 2023

## Miscellaneous

- Citizenship: United States of America
- Languages: English (native), Japanese (basic)